Internship Possible Questions

1. Most important questions

These are the questions I would prepare first:

1. Tell me about your internship at Bluestock Fintech.
2. What exactly was your role as a Data Analyst Intern?
3. Explain the project/workflow you worked on.
4. Explain your complete ETL pipeline.
5. Why did you use Python for ETL?
6. Why Pandas?
7. Why SQLAlchemy?
8. Why SQLite?
9. How did you integrate the mfapi.in REST API?
10. What is NAV?
11. What is AMFI?
12. How did you handle missing data?
13. What data anomalies did you encounter?
14. How did you validate the quality of the data?
15. How did you process 40 mutual funds?
16. What analysis did you perform?
17. Why did you use Power BI?
18. What DAX measures did you create?
19. What is cross-filtering?
20. What is drill-through?

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2. "Tell me about your internship"

This is almost guaranteed.

You should be able to explain it as:

Raw financial data

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Python / Pandas

↓

Data Cleaning

↓

Data Transformation

↓

SQLAlchemy

↓

SQLite

↓

Analysis

↓

Power BI

↓

Interactive Dashboard

And separately:

mfapi.in REST API

↓

Fetch live NAV data

↓

Python

↓

Clean / transform

↓

Database

↓

Analysis / Dashboard

Your resume explicitly mentions both the ETL pipeline and live NAV API ingestion.

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3. ETL questions — VERY IMPORTANT

Because **ETL pipeline** is one of your strongest resume claims, expect detailed questioning.

21. What is ETL?
22. Explain Extract, Transform and Load.
23. Where was the data extracted from?
24. What transformations did you perform?
25. Where did you load the data?
26. Why did you use an ETL pipeline?
27. Why not clean the data manually?
28. How did you automate the pipeline?
29. How did you handle errors during extraction?
30. How did you handle invalid records?
31. How did you ensure duplicate data wasn't inserted?
32. How did you validate the transformed data?
33. What happens if the source API is unavailable?
34. How would you make the pipeline scalable?
35. How would you schedule the ETL process?
36. What is the difference between ETL and ELT?
37. What is batch processing?
38. What is real-time data processing?
39. Is your pipeline batch or real-time?
40. How would you modify it to support real-time ingestion?

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4. Pandas questions

Your resume specifically mentions Pandas for ETL and processing.

41. What is Pandas?
42. What is a DataFrame?
43. Series vs DataFrame?
44. How do you read a CSV using Pandas?
45. How do you read JSON data?
46. How do you handle missing values?
47. Difference between dropna() and fillna()?

48. How do you remove duplicate records?
49. How do you filter rows?
50. How do you group data?
51. What is groupby()?
52. Difference between merge(), join(), and concat()?
53. How do you sort a DataFrame?
54. How do you detect outliers?
55. How do you change a column's datatype?
56. What is vectorization?
57. Why is Pandas faster than traditional Python loops for many operations?
58. How do you handle a very large dataset that doesn't fit into memory?

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5. NumPy questions

Since you explicitly mention **Pandas and NumPy** for benchmarking 40 funds:

59. What is NumPy?
60. Why use NumPy instead of Python lists?
61. What is an ndarray?
62. What is vectorization?
63. What statistical operations did you perform using NumPy?
64. What is mean?
65. What is median?
66. Mean vs median?
67. What is standard deviation?
68. What is variance?
69. How did you compare the 40 mutual funds?
70. Which metrics did you use for comparison?

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6. SQLAlchemy questions

You have specifically listed **SQLAlchemy**, which makes it fair game.

71. What is SQLAlchemy?
72. Why did you use SQLAlchemy?
73. SQLAlchemy vs raw SQL?
74. What is ORM?
75. What is a database engine?
76. How does Python connect to SQLite?
77. How did you create tables?
78. How did you insert data?
79. How did you query the database?
80. How did you handle transactions?
81. What happens if an insertion fails?
82. Why not use SQLite's Python library directly?

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7. SQLite questions

83. Why SQLite?
84. What is SQLite?

85. Is SQLite a server-based database?
86. SQLite vs PostgreSQL?
87. SQLite vs MySQL?
88. What are SQLite's limitations?
89. When is SQLite a good choice?
90. Can SQLite handle concurrent users?
91. Why would you migrate from SQLite to PostgreSQL?

A strong answer should connect the choice to the **scale and purpose of the internship pipeline**, rather than claiming SQLite is universally better.

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8. REST API / mfapi.in questions

Your resume specifically states that you integrated **mfapi.in REST API** to automate live NAV data ingestion.

92. What is a REST API?
93. Why use REST APIs?
94. How does a REST API work?
95. What HTTP methods do you know?
96. What is GET?
97. What is POST?
98. What is an endpoint?
99. What is JSON?
100. How did you call the mfapi.in API?
101. What response format did you receive?
102. How did you parse the API response?
103. How did you handle API errors?
104. What happens if the API returns HTTP 404?
105. What happens if it returns 500?
106. What happens if the API times out?
107. How did you handle rate limiting?
108. How did you prevent duplicate NAV records?
109. How did you automate recurring data ingestion?

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9. Mutual fund / finance questions

Because this was a **fintech internship**, don't assume they'll only ask technical questions.

110. What is a mutual fund?
111. What is NAV?
112. How is NAV calculated?
113. What is an AMC?
114. What is AMFI?
115. What is an equity mutual fund?
116. What is a debt mutual fund?
117. What is a hybrid mutual fund?
118. What is SIP?
119. SIP vs lump-sum investment?
120. What is the difference between NAV and fund return?
121. Can a lower NAV mean a fund is cheaper?

122. What factors affect mutual fund performance?
123. What does benchmark mean in mutual funds?
124. What metrics would you use to compare mutual funds?

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10. Data cleaning questions

This is particularly important because your resume says you **resolved missing data anomalies**.

125. What types of missing data did you encounter?
126. How did you identify missing values?
127. How did you decide whether to remove or replace them?
128. When would you use mean imputation?
129. When would you use median?
130. What are duplicate records?
131. How did you detect duplicates?
132. What are outliers?
133. How did you identify outliers?
134. What is an anomaly?
135. How did you verify that an anomaly wasn't actually valid data?
136. How did you ensure cleaning didn't alter legitimate information?
137. How did you maintain data consistency?

Very likely follow-up:

"Give me one real example of a data anomaly you encountered and exactly how you handled it."

You should have a **specific real example from your internship** ready rather than giving a textbook definition.

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11. Statistical analysis questions

Your resume mentions **statistical summaries using Matplotlib/Seaborn**.

138. What is descriptive statistics?
139. What statistical measures did you calculate?
140. What is mean?
141. What is median?
142. What is standard deviation?
143. What is percentile?
144. What is correlation?
145. Correlation vs causation?
146. What visualization did you use?
147. Why did you choose a particular chart?
148. How did visualization help your analysis?
149. How did you identify trends in mutual fund performance?

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12. Power BI questions — HIGH PRIORITY

Your resume says you designed **interactive Power BI dashboards with custom DAX, cross-filtering and drill-through**.

Expect:

150. What is Power BI?
151. Why Power BI?

152. What is Power Query?
153. Power Query vs DAX?
154. What is DAX?
155. Why do we need DAX?
156. What is a measure?
157. What is a calculated column?
158. Measure vs calculated column?
159. What DAX measures did you create?
160. What is CALCULATE()?
161. What is filter context?
162. What is row context?
163. What is context transition?
164. What is a slicer?
165. What is cross-filtering?
166. What is drill-through?
167. Drill-through vs drill-down?
168. How did you design your dashboard?
169. How did you make the dashboard interactive?
170. How did you improve dashboard performance?

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13. DAX questions — prepare properly

If you write **"custom DAX measures"** on your resume, an interviewer may say:

"Give me an example of a DAX measure you created."

Then:

171. What is a DAX measure?
172. Why not calculate the value in Python?
173. What is SUM()?
174. What is AVERAGE()?
175. What is CALCULATE()?
176. What is FILTER()?
177. What is DIVIDE()?
178. What is filter context?
179. What is row context?
180. How does CALCULATE() modify filter context?

You should be able to explain a real measure from your actual dashboard.

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14. Dashboard questions

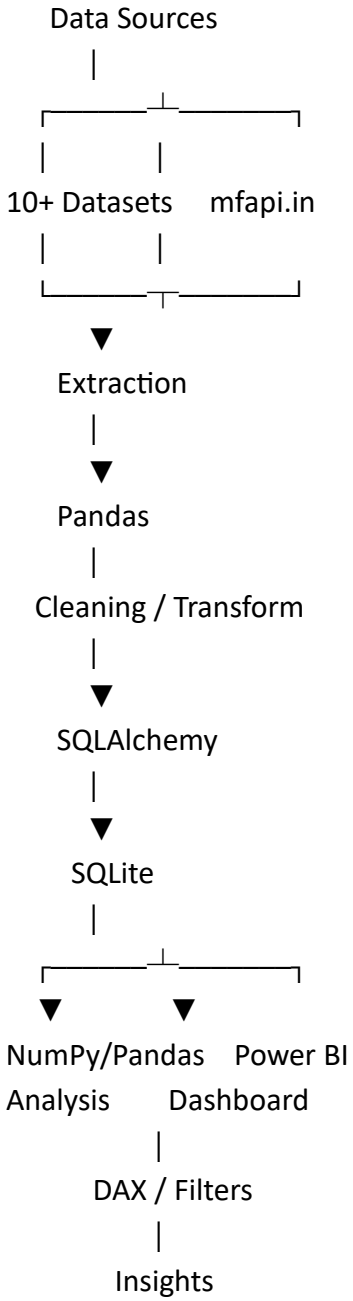
181. What KPIs did you show?
182. Why did you choose those KPIs?
183. What charts did you use?
184. Why did you choose those particular visualizations?
185. How did users interact with the dashboard?
186. How did cross-filtering work?
187. How did drill-through work?
188. How did you handle large datasets in Power BI?

189. How did you ensure the dashboard was accurate?
190. How did you validate the dashboard against the source database?

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15. Data pipeline architecture questions

An interviewer may ask you to draw the entire internship architecture:



Then expect:

191. Why was this architecture chosen?
192. Where was the bottleneck?
193. Which component was responsible for cleaning?
194. Which component performed analysis?
195. Which component stored the data?
196. How did you ensure consistency between API data and dashboard data?
197. How would you productionize this pipeline?

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16. Questions about scale

Because your resume says **10+ datasets** and **40 mutual funds**, interviewers may probe scalability.

198. What happens if the number of funds grows from 40 to 40,000?
199. What happens if the datasets grow from 10 to 1,000?
200. Would SQLite still be appropriate?
201. Would Pandas still be appropriate?
202. How would you optimize the pipeline?
203. Would you process data in batches?
204. How would you parallelize API calls?
205. How would you handle API rate limits?
206. How would you schedule the pipeline?
207. How would you monitor failures?

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17. Troubleshooting questions

These are excellent interview questions because they test whether you actually built the system:

208. The API suddenly returns no data. What do you check?
209. The database contains duplicate NAV entries. How do you debug it?
210. Pandas shows unexpected null values. What do you do?
211. The dashboard shows a different number from the database. How do you investigate?
212. Your API ingestion becomes slow. How do you optimize it?
213. The Power BI dashboard is slow. What would you check?
214. A mutual fund's NAV suddenly looks incorrect. How would you validate it?
215. What logging did you use?
216. How did you handle exceptions?

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18. Behavioral questions about the internship

Don't prepare only technical questions.

217. What was your biggest learning from the internship?
218. What was the most difficult problem you faced?
219. Tell me about a bug you encountered.
220. How did you solve that bug?
221. Did you work independently or with a team?
222. How did you communicate your findings?
223. What did you learn about financial data?
224. What would you do differently now?
225. How did this internship influence your career interests?
226. What was your most significant contribution?

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19. Top 25 questions to master

If you have limited time, these are the ones I'd prioritize:

Priority	Question
🔴 1	Tell me about your internship
🔴 2	What exactly was your role?
🔴 3	Explain your complete ETL pipeline
🔴 4	What is ETL?

Priority		Question
🔴	5	What data sources did you use?
🔴	6	Why Python/Pandas?
🔴	7	Why SQLAlchemy?
🔴	8	Why SQLite?
🔴	9	Explain mfapi.in integration
🔴	10	What is REST API?
🔴	11	What is NAV?
🔴	12	What is AMFI?
🔴	13	How did you handle missing data?
🔴	14	Give an example of a data anomaly
🔴	15	How did you validate data quality?
🔴	16	How did you analyze 40 mutual funds?
🔴	17	What statistical measures did you use?
🔴	18	Why Power BI?
🔴	19	What is DAX?
🔴	20	Measure vs calculated column
🔴	21	What DAX measures did you create?
🔴	22	What is cross-filtering?
🔴	23	What is drill-through?
🔴	24	How did you validate your dashboard?
🔴	25	What would you improve in the pipeline?

The biggest interview rule for this internship

There are **three numbers/claims on your resume that can trigger deeper questioning**:

"10+ datasets" → *What datasets? What format? How did you process them?*

"40 mutual funds" → *Which metrics? How did you benchmark them? What did you discover?*

"live NAV data ingestion" → *Show me the API flow. What endpoint? What response? How did you handle errors/duplicates?*